

Section 1. Chemical product and company identification

Chinese name : 殺菌洗手液
Product name English : DIGICLEAN E HAND SOAP
Supplier : Ecolab Limited
 15/F, Lu Plaza, 2 Wing Yip Street, Kwun Tong, Kowloon, HONG KONG
Emergency telephone number : 852-2341-4202
Product type : Skin Cleaner
Date of issue : 2009 . 10 . 26

Section 2. Hazards identification

Hazard classification : Not available.
Routes of entry : Skin contact, Eye contact, Inhalation, Ingestion
Human health hazards
Inhalation : Slightly irritating to the respiratory system.
Ingestion : No known significant effects or critical hazards.
Skin : No known significant effects or critical hazards.
Eyes : Moderately irritating to eyes.
Environmental hazards : Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.
Fire/explosion hazards : Combustible liquid. In a fire or if heated, a pressure increase will occur and the container may burst, with the risk of a subsequent explosion.

Section 3. Composition/information on ingredients

Mixture

Chemical name	Concentration	CAS No.
ethanol	1 - 10	64-17-5
poly(oxy-1,2-ethanediyl), .alpha.-sulfo-.omega.-hydroxy-, c10-16-alkyl ethers, sodium salts	1 - 10	68585-34-2
2-methylpentane-2,4-diol	1 - 10	107-41-5
boric acid	<1.0	10043-35-3

Section 4. First aid measures

Inhalation : If inhaled, remove to fresh air.
Ingestion : Do not induce vomiting. Never give anything by mouth to an unconscious person. If irritation persists, get medical attention.
Skin contact : Flush contaminated skin with plenty of water. Wash clothing before reuse.
Eye contact : In case of contact, immediately flush eyes with cool running water. Remove contact lenses and continue flushing with plenty of water for at least 15 minutes. Get medical attention if irritation persists.

Section 5. Fire-fighting measures

Extinguishing media

Suitable	: Use dry chemical, CO ₂ , water spray (fog) or foam.
Not suitable	: Do not use water jet.
Hazardous combustion products	: Decomposition products may include the following materials: carbon dioxide carbon monoxide metal oxide/oxides
Special protective equipment for fire-fighters	: Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.
Materials to avoid	None identified.

Section 6. Accidental release measures

Personal precautions	: Ventilate area of leak or spill. Do not touch damaged containers or spilled material unless wearing appropriate protective equipment (Section 8). Stop leak if without risk. Do not allow to enter drains or watercourses.
Environmental precautions	: Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).
Methods for cleaning up	: If emergency personnel are unavailable, contain spilled material. For small spills, add absorbent (soil may be used in the absence of other suitable materials) and use a non-sparking or explosion-proof means to transfer material to a sealable, appropriate container for disposal. For large spills, dike spilled material or otherwise contain it to ensure runoff does not reach a waterway. Place spilled material in an appropriate container for disposal.

Section 7. Handling and storage

Handling	: Avoid contact with eyes. Keep away from heat, sparks and flame. To avoid fire, eliminate ignition sources. Wash thoroughly after handling.
Storage	: Keep out of reach of children. Store in a segregated and approved area. Keep container in a cool, well-ventilated area. Keep container tightly closed and sealed until ready for use. Avoid all possible sources of ignition (spark or flame). Do not store above the following temperature: 50°C

Section 8. Exposure controls/personal protection

Chemical name	Exposure limits
ethanol	ACGIH TLV (United States, 1/2009). STEL: 1000 ppm 15 minute(s).
2-methylpentane-2,4-diol	GBZ-2 (China, 4/2007). MAC: 100 mg/m ³

Recommended monitoring procedures	: If this product contains ingredients with exposure limits, personal, workplace atmosphere or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity to use respiratory protective equipment.
Occupational exposure controls	: Use only with adequate ventilation. Use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits. The engineering controls also need to keep gas, vapor or dust concentrations below any lower explosive limits. Use explosion-proof ventilation equipment.
Hygiene measures	: Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing. Ensure that eyewash stations and safety showers are close to the workstation location.

Personal protection

Section 8. Exposure controls/personal protection

Respiratory	A respirator is not needed under normal and intended conditions of product use.
Hands	: No protective equipment is needed under normal use conditions.
Eyes	: Eye protection recommended.
Skin	: No protective equipment is needed under normal use conditions.

Section 9. Physical and chemical properties

Physical state	: Liquid. [Gel]
Color	: Blue.
Odor	: citrus
pH	: 8.75 [Conc. (% w/w): 100%]
Flash point	: 82.2222 °C (Closed cup)
Specific gravity	: 1.02

Section 10. Stability and reactivity

Stability and reactivity	: The product is stable.
Conditions to avoid	: Avoid all possible sources of ignition (spark or flame). Do not pressurize, cut, weld, braze, solder, drill, grind or expose containers to heat or sources of ignition.
Materials to avoid	: None identified.
Hazardous decomposition products	: Under normal conditions of storage and use, hazardous decomposition products should not be produced.

Section 11. Toxicological information**Acute toxicity**

Inhalation	: Slightly irritating to the respiratory system.
Ingestion	: No known significant effects or critical hazards.
Skin	: No known significant effects or critical hazards.
Eyes	: Moderately irritating to eyes.

Potential chronic health effects

Chronic effects	: No known significant effects or critical hazards.
Carcinogenicity	: No known significant effects or critical hazards.
Mutagenicity	: No known significant effects or critical hazards.
Teratogenicity	: No known significant effects or critical hazards.
Developmental effects	: No known significant effects or critical hazards.
Fertility effects	: No known significant effects or critical hazards.

Section 12. Ecological information**Aquatic ecotoxicity**

Product/ingredient name	Result	Species	Exposure
2-methylpentane-2,4-diol	Acute EC50 3300 mg/L	Daphnia	48 hours
	Acute EC50 3200 mg/L	Daphnia	48 hours
	Acute LC50 10700 mg/L	Fish	96 hours
	Acute LC50 >10000 mg/L	Fish	96 hours
ethanol	Acute LC50 13000 mg/L	Fish	96 hours
	Acute LC50 >100 mg/L	Fish	96 hours
boric acid	Acute EC50 777 to 932 ppm	Daphnia - Water flea -	48 hours
	Fresh water	Daphnia magna - <24 hours	
	Acute EC50 226 to 246 ppm	Daphnia - Water flea -	48 hours
	Fresh water	Daphnia magna - <24 hours	
	Acute EC50 133 to 153 ppm	Daphnia - Water flea -	48 hours

Section 12. Ecological information

hours

Acute LC50 >1100 ppm Fresh water	Fish - Rainbow trout,donaldson trout - Oncorhynchus mykiss	96 hours
Acute LC50 >1021 ppm Fresh water	Fish - Bluegill - Lepomis macrochirus	96 hours
Acute LC50 >800 ppm Fresh water	Fish - Rainbow trout,donaldson trout - Oncorhynchus mykiss	96 hours
Acute LC50 92.83 to 148 mg/L Marine water	Crustaceans - Opossum shrimp - Americamysis bahia - Juvenile (Fledgling, Hatchling, Weanling) -	48 hours
Acute LC50 89.07 to 100.7 mg/L Marine water	Crustaceans - Opossum shrimp - Americamysis bahia - Juvenile (Fledgling, Hatchling, Weanling) -	48 hours
Acute LC50 5600 ppm Fresh water	Fish - Western mosquitofish - Gambusia affinis - Adult	96 hours
Acute LC50 50 to 100 ppm Fresh water	Fish - Rainbow trout,donaldson trout - Oncorhynchus mykiss	96 hours
Acute LC50 >100000 ug/L Fresh water	Fish - Colorado squawfish - Ptychocheilus lucius - Juvenile (Fledgling, Hatchling, Weanling) -	96 hours
Acute LC50 233000 to 293000 ug/L Fresh water	Fish - Razorback sucker - Xyrauchen texanus - Swim-up - 10 to 17 days	96 hours
Acute LC50 125000 to 162000 ug/L Fresh water	Fish - Flannelmouth sucker - Catostomus latipinnis - LARVAE - 12 to 13 days	96 hours

Other adverse effects : No known significant effects or critical hazards.

Environmental effects : Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

Section 13. Disposal considerations

Waste disposal : The generation of waste should be avoided or minimized wherever possible. Empty containers or liners may retain some product residues. This material and its container must be disposed of in a safe way. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

Consult your local or regional authorities.

Section 14. Transport information

Package mark : Not regulated.

Packing group : -

Section 15. Regulatory information

China inventory (IECSC) : All components are listed or exempted.

Authored with regard to:

- "Classification and Mark for commonly used dangerous chemicals" (GB13690-1992)
- "General rules for preparation of a Chemical Safety Data Sheet" (GB16483-2000)
- "The principle of classification of transport packaging groups of dangerous goods" (GB/T15098-94)
- "Classification and code of dangerous goods" (GB6944-2005)

Section 16. Other information

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Prepared by : Regulatory Affairs

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☑ Indicates information that has changed from previously issued version.

Notice to reader

The above information is believed to be correct with respect to the formula used to manufacture the product in the country of origin. As data, standards, and regulations change, and conditions of use and handling are beyond our control, NO WARRANTY, EXPRESS OR IMPLIED, IS MADE AS TO THE COMPLETENESS OR CONTINUING ACCURACY OF THIS INFORMATION.